

2a objective 1

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clc;
clear;
close all;

%% Input Parameters
Am = 2;          % Message signal amplitude
fm = 500;        % Message frequency (Hz)
Ac = 1;          % Carrier amplitude
fc = 10000;      % Carrier frequency (Hz)
kf = 1000;       % Frequency sensitivity (Hz/V)
fs = 100000;     % Sampling frequency (Hz)
T = 0.02;        % Simulation duration (s)
t = 0:1/fs:T;

%% Message Signal
m = Am*cos(2*pi*fm*t);

%% Frequency Deviation Calculation
delta_f = kf * Am;

%% Modulation Index Calculation
beta = delta_f / fm

%% FM Signal Generation
fm_signal = Ac*cos(2*pi*fc*t + beta*sin(2*pi*fm*t));

%% Plot Message Signal
figure;
subplot(2,1,1);
plot(t,m,'LineWidth',1.5);
title('Message Signal');
xlabel('Time (s)');
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ylabel('Amplitude');
grid on;
%% Plot FM Signal
subplot(2,1,2);
plot(t,fm_signal,'LineWidth',1.5);
title('Frequency Modulated Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
%% Spectrum Analysis
N = length(fm_signal);
FM_fft = fftshift(abs(fft(fm_signal))/N);
f = (-N/2:N/2-1)*(fs/N);
figure;
plot(f,FM_fft,'LineWidth',1.5);
title('FM Signal Spectrum');
xlabel('Frequency (Hz)');
ylabel('Magnitude');
grid on;
xlim([fc-10000 fc+10000]);
%% Display Results
fprintf('\nFM SIGNAL PARAMETERS\n');
fprintf('-----\n');
fprintf('Carrier Frequency    = %.0f Hz\n', fc);
fprintf('Message Frequency     = %.0f Hz\n', fm);
fprintf('Message Amplitude      = %.2f V\n', Am);
fprintf('Frequency Sensitivity  = %.0f Hz/V\n', kf);
fprintf('\nCALCULATED VALUES\n');
fprintf('-----\n');

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fprintf('Frequency Deviation ( $\Delta f$ ) = %.0f Hz\n', delta_f);
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fprintf('Modulation Index ( $\beta$ ) = %.2f\n', beta);
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%% Carson's Rule Bandwidth
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BW = 2*(delta_f + fm);
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fprintf('Bandwidth (Carson Rule) = %.0f Hz\n', BW);
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